Rigidity, Form, and Compactness of Invertible a.e. Measure-Preserving Maps of Bounded Lipschitz Domains in $\mathbb{R}^n$

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1 Basic Properties

Definition 1 *Throughout the paper, $\Omega \subset \mathbb{R}^n$ always denotes a bounded Lipschitz Domain with $0 < \mu(\Omega) < \infty$ (for μ the usual Lebesgue measure).*

Definition 2 *By convention we take the range of functions in $L^p(\Omega)$ to $\mathbb{R}^n$. If the range is instead K, it will be denoted as $L^p(\Omega, K)$.*

Definition 3 *We let $\sigma(\Omega)$ be the set of invertible a.e. measure preserving maps $\Omega \to \Omega$ which are locally linear a.e. with a unitary Jacobian*

$\sigma(\Omega)$ also has a group structure as well.

Proposition 1 *For $\tau, \gamma \in \sigma(\Omega)$ we have that $\tau \circ \gamma \in \sigma(\Omega)$.*

Proposition 2 *For all $\tau \in \sigma(\Omega)$, there exists some function g such that $\tau \circ g = \mathrm{Id}$ almost everywhere. In addition, $g \in \sigma(\Omega)$ as well. g is usually denoted as τ^{-1} even though the actual inverse is not explicitly defined and need not exist.*

2 Rigidity and Form

For any $\tau \in \sigma(\Omega)$, because it is locally linear a.e. we have that there exists some set $S \subset \Omega$ of full measure such that for every $x \in S$ we have that there is a ball $B_{r_x}(x) \subset \Omega$ on which τ is equivalent to a linear map $Ux + v$. We let $S(\tau) = \cup_{x \in S} B_{r_x}(x)$ be the open set of all points in Ω at which τ is locally linear. Note that $\mu(S(\tau)) \geq \mu(S) = \mu(\Omega)$ so that $S(\tau)$ is also a full subset.

Definition 4 *We take $P_{S(\tau)}$ to be the set of connected components of $S(\tau)$.*

On each $C \in P_{S(\tau)}$ we have that C is open so that τ is equal a.e. to $Ux + v$ for unitary U on C. We define the function $g_\tau : S(\tau) \to \Omega$ by $g_\tau(x) = Ux + v$ for $x \in C$. This is equivalent a.e. to τ but is continuous a.e. as well. We may extend g_τ to Ω such that $g_\tau(\Omega) = \Omega$. Note that $S(\tau^{-1}) = g_\tau(S(\tau))$.

Proposition 3 *For any a.e. invertible L^p function $f : \Omega \to \Omega$, we have that $f \in \sigma(\Omega)$ if and only if for almost all $x \in \Omega$ there exists some $\varepsilon > 0$ such that there are $U \in U(n)$ and $v \in \mathbb{R}^n$ such that $f(y) \equiv Uy + v$ for $y \in B(x, \varepsilon)$.*

Proof. The $\implies$ direction follows from the definition. Now, conversely, assume that for all $x \in S$ there exists some $\varepsilon > 0$ such that $f(y) \equiv Uy + v$ for $y \in B(x, \varepsilon)$, where $\mu(S) = \mu(\Omega)$.
Now, for all $y \in f(S)$ there exists some $\varepsilon > 0$ such that $f^{-1}(z) \equiv U^T z - U^T v$ for all $z \in B(y, \varepsilon)$. Then,

$(f^{-1})'(y)$ is unitary.

$$\begin{aligned}\mu(f(A)) &= \int_\Omega 1_A \circ f^{-1} d\mu \\ &= \int_S 1_A \circ f^{-1} d\mu \\ &= \int_{f(S)} 1_A |\det(f^{-1})'| d\mu \\ &= \int_\Omega 1_A d\mu \\ &= \mu(A)\end{aligned}$$

And therefore, $f \in \sigma(\Omega)$.

Proposition 4 *For $\tau, \gamma \in \sigma(\Omega)$, we have that $S(\tau \circ \gamma) \subseteq \Omega \setminus (S(\gamma) \Delta g_\gamma^{-1}(S(\tau)))$.*

Proof. Assume $x \in S(\tau \circ \gamma)$. If g_γ is continuous at x we have that g_τ is continuous at $g_\gamma(x)$ so $x \in S(\gamma) \cap g_\gamma^{-1}(S(\tau))$.
Otherwise, if $x \notin S(\gamma)$ then if $g_\gamma(x) \in S(\tau)$ we have that $g_\tau(g_\gamma(x))$ cannot be continuous, a contradiction. Therefore, $x \in (\Omega \setminus S(\gamma)) \cap (\Omega \setminus g_\gamma^{-1}(S(\tau))) = \Omega \setminus (S(\gamma) \cup g_\gamma^{-1}(S(\tau)))$.
Finally, $S(\tau \circ \gamma) \subseteq (\Omega \setminus (S(\gamma) \cup g_\gamma^{-1}(S(\tau)))) \cup (S(\gamma) \cap g_\gamma^{-1}(S(\tau))) = \Omega \setminus (S(\gamma) \Delta g_\gamma^{-1}(S(\tau)))$.
Corollary. $S(\gamma) \cap g_\gamma^{-1}(S(\tau)) \subseteq S(\tau \circ \gamma)$

Proposition 5 *For all $\tau \in \sigma(\Omega)$ and $X \subset \Omega$, we have that $H^1(X) = H^1(\tau(X))$.*

Proof. TODO

3 Estimates on Measures of Thickened Sets of Discontinuity

We now wish to study the topological and compactness properties of $\sigma(\Omega)$, treating it as a subset of L^p. With the Rellich Kondrachov theorem, we see that compactness of sets in L^p is in some sense correlated to the uniform convergence of $||\tau(x+h) - \tau(x)||_{L^p(\Omega)} \to 0$ as $|h| \to 0$. For h sufficiently small so that it $B(x,h)$ contains no discontinuities of g_τ, $|g_\tau(x+h) - g_\tau(x)| = |h|$, so the uniformity is almost entirely determined by the size of the set of discontinuities of g_τ. Thus, in order to study compactness, we first study the size of the sets of discontinuity of each τ.

Definition 5 *We define $A(\tau, \varepsilon) = \{x \in \Omega \mid d(x, \partial(\Omega \setminus S(\tau)) < \varepsilon\}$.*

Note that $\partial(\Omega \setminus S(\tau)) = \Omega \setminus S(\tau)$, and are interchanged at times.

Lemma 1 *For any open set X with $X \subseteq S(\tau)$ and $\mu(X) = \mu(\Omega)$, we have that $\mu(\{x \in \Omega \mid d(x, \Omega \setminus g_\tau(X)) < \varepsilon\}) = \mu(\{x \in \Omega \mid d(x, \Omega \setminus X) < \varepsilon\})$*

Proof. We have that $\mu(\{x \in \Omega \mid d(x, \Omega \setminus g_\tau(X)) < \varepsilon\}) = \mu(\{x \in g_\tau(X) \mid d(x, \Omega \setminus g_\tau(X)) < \varepsilon\})$.
Now, assume for any $x \in g_\tau(X)$ that $y \in \Omega \setminus g_\tau(X)$ minimizes $|x - y|$. We see that $x \in C$ where C is taken to be the largest (by inclusion) convex connected component of $g_\tau(X)$ containing x. Then, $y \in \partial C$ by the minimality of $|x - y|$.
In addition, we have that $g_\tau(X) \subseteq g_\tau(S(\tau)) = S(\tau^{-1})$ so that $g_\tau^{-1}(x) = Ux + v$ on C.
Now, with $\ell = \{x + a(y - x) \mid a \in [0, 1)\}$, we have that $\ell \subset C$ by the convexity of ∂C (because of the convexity of C).
Thus, $|x-y| = H^1(\ell) = H^1(g_\tau^{-1}(\ell)) \geq |g_\tau^{-1}(x) - y_2|$ for some $y_2 \in \partial g_\tau^{-1}(C)$. Thus, $|x-y| \geq d(g_\tau^{-1}(x), g_\tau^{-1}(C)) \geq d(g_\tau^{-1}(x), \Omega \setminus g_\tau^{-1}(g_\tau(X))) = d(g_\tau^{-1}(x), \Omega \setminus X)$.
Finally, we have the following.

$$\begin{aligned}\implies \mu(\{x \in \Omega \mid d(x, \Omega \setminus g_\tau(X)) < \varepsilon\}) &\leq \mu(\{x \in g_\tau(X) \mid d(g_\tau^{-1}(x), \Omega \setminus X) < \varepsilon\}) \\ &= \mu(\{x \in \Omega \mid d(x, \Omega \setminus X) < \varepsilon\})\end{aligned}$$

Now, as $X \subseteq S(\tau)$ we see that $g_\tau(X) \subseteq S(\tau^{-1})$ so that $\mu(\{x \in \Omega \mid d(x, \Omega \setminus X) < \varepsilon\}) = \mu(\{x \in \Omega \mid d(x, \Omega \setminus g_\tau^{-1}(g_\tau(X))) < \varepsilon\}) \leq \mu(x \in \Omega \mid d(x, \Omega \setminus g_\tau(X)) < \varepsilon\})$.
Therefore, $\mu(\{x \in \Omega \mid d(x, \Omega \setminus X) < \varepsilon\}) = \mu(\{x \in \Omega \mid d(x, \Omega \setminus g_\tau(X)) < \varepsilon\})$.

Proposition 6 *$\mu(A(\tau \circ \gamma^{-1}, \varepsilon)) \leq \mu(A(\tau, \varepsilon)) + \mu(A(\gamma, \varepsilon))$ for all $\varepsilon > 0$ and $\tau, \gamma \in \sigma(\Omega)$.*

Proof.

$$\begin{aligned}\mu(A(\tau \circ \gamma^{-1}, \varepsilon)) &= \mu(\{x \in \Omega \mid d(x, \Omega \setminus S(\tau \circ \gamma^{-1})) < \varepsilon\}) \\ &\leq \mu(\{x \in \Omega \mid d(x, \Omega \setminus (S(\gamma^{-1}) \cap g_\gamma(S(\tau)))) < \varepsilon\}) \\ &\leq \mu(\{x \in \Omega \mid d(x, \Omega \setminus (S(\gamma^{-1}) \cap g_\gamma(S(\tau) \cap S(\gamma)))) < \varepsilon\}) \\ &= \mu(\{x \in \Omega \mid d(x, \Omega \setminus g_\gamma(S(\tau) \cap S(\gamma))) < \varepsilon\})\end{aligned}$$

By Lemma 1 we have that $\mu(\{x \in \Omega \mid d(x, \Omega \setminus g_\gamma(S(\tau) \cap S(\gamma))) < \varepsilon\}) = \mu(\{x \in \Omega \mid d(x, (S(\tau) \cap S(\gamma))) < \varepsilon\})$ because $S(\tau) \cap S(\gamma) \subseteq S(\gamma)$ and has full measure.

$$\begin{aligned}\implies \mu(A(\tau \circ \gamma^{-1}, \varepsilon)) &\leq \mu(\{x \in \Omega \mid d(x, \Omega \setminus (S(\tau) \cap S(\gamma))) < \varepsilon\}) \\ &\leq \mu(\{x \in \Omega \mid d(x, \Omega \setminus S(\tau)) < \varepsilon\}) + \mu(\{x \in \Omega \mid d(x, \Omega \setminus S(\gamma)) < \varepsilon\}) \\ &= \mu(A(\tau, \varepsilon)) + \mu(A(\gamma, \varepsilon))\end{aligned}$$

Intuitively, we expect that $A(\gamma, \varepsilon)$ and $A(\gamma^{-1}, \varepsilon)$ are very similar because $S(\gamma^{-1}) = g_\gamma(S(\gamma))$ and because $g_\gamma \in \sigma(\Omega)$ are locally unitary and invertible, they tend to preserve the geometry of sets in some sense.

Proposition 7 *We have that $\mu(A(\gamma, \varepsilon)) = \mu(A(\gamma^{-1}, \varepsilon))$ for all $\gamma \in \sigma(\Omega)$ and $\varepsilon > 0$.*

Proof. We have the following estimate.

$$\begin{aligned}\mu(A(\gamma^{-1}, \varepsilon)) &= \mu(\{x \in \Omega \mid d(x, \Omega \setminus S(\gamma^{-1})) < \varepsilon\}) \\ &= \mu(\{x \in \Omega \mid d(x, \Omega \setminus g_\gamma(S(\gamma)) < \varepsilon\})\end{aligned}$$

We see that $S(\gamma) \subseteq S(\gamma)$ so by Lemma 1 we have that $\mu(A(\gamma^{-1}, \varepsilon)) = \mu(\{x \in \Omega \mid d(x, \Omega \setminus S(\gamma)) < \varepsilon\}) = \mu(A(\gamma, \varepsilon))$.

4 Topological Properties

We are now ready to study the topological properties of $\sigma(\Omega)$.

Proposition 8 *If $X \subseteq \sigma(\Omega)$ is open under $||\cdot||_{L^p(\Omega)}$, then $X \circ \tau$ is as well for all $\tau \in \sigma(\Omega)$.*

Proof. We have that for any $\gamma \in X \circ \tau$ that $\gamma \circ \tau^{-1} \in X$ and thus there is some $B = B_{L^p(\Omega)}(\gamma \circ \tau^{-1}, \varepsilon) \subset X$. For $\lambda \in B$ we have that $||\lambda \circ \tau - \gamma||_{L^p(\Omega, \mathbb{R}^n)} = ||\lambda - \gamma \circ \tau^{-1}||_{L^p(\Omega)}$, so that $B_{L^p(\Omega)}(\gamma \circ \tau^{-1}, \varepsilon) \circ \tau = B_{L^p(\Omega)}(\gamma, \varepsilon) \subset X \circ \tau$. And thus there is an open neighborhood of $\gamma \in X \circ \tau$, so that $X \circ \tau$ is open as well.

Next, for $\tau, \gamma, \lambda \in \sigma(\Omega)$ we see that $||\tau \circ \lambda - \gamma \circ \lambda||_{L^p(\Omega)} = ||\tau - \gamma||_{L^p(\Omega)}$ so that convergence in $||\cdot||_{L^p(\Omega)}$ is right-invariant. This raises the question of whether or not convergence is left-invariant as well. We expect that because distances $|\lambda(x) - \lambda(y)|$ are preserved (a.e) if $|x - y|$ is small enough and $||\tau - \gamma||_{L^p(\Omega)}$ being small forces $|\tau - \gamma|$ to be small as well that $|\lambda \circ \tau - \lambda \circ \gamma|$ must be small as well, constrained by $||\tau - \gamma||_{L^p(\Omega)}$.

Proposition 9 *If $||\gamma_n - \gamma||_{L^p(\Omega)} \to 0$, then $||\tau \circ \gamma_n - \tau \circ \gamma||_{L^p(\Omega)} \to 0$ as well for all $\tau \in \sigma(\Omega)$.*

Proof. Note that $\mu(E_\delta) = \mu(\{x \in \Omega \mid |\gamma_n(x) - \gamma(x)| > \delta\}) < \delta^{-p}||\gamma_n(x) - \gamma(x)||_{L^p(\Omega)}$ by Chebyshev's inequality.

$$||\tau \circ \gamma_n - \tau \circ \gamma||^p_{L^p(\Omega)} = \int_\Omega |\tau \circ \gamma_n - \tau \circ \gamma|^p d\mu$$
$$= \int_{E_\delta} |\tau \circ \gamma_n - \tau \circ \gamma|^p d\mu + \int_{\Omega \setminus E_\delta} |\tau \circ \gamma_n - \tau \circ \gamma|^p d\mu$$
$$\leq \delta^{-p}||\gamma_n - \gamma||_{L^p} \operatorname{diam}(\Omega)^p + I$$
$$I = \int_{(\Omega \setminus E_\delta) \setminus g_\gamma^{-1}(A(\tau,2\delta))} |\tau \circ \gamma_n - \tau \circ \gamma|^p d\mu + \int_{(\Omega \setminus E_\delta) \cap g_\gamma^{-1}(A(\tau,2\delta))} |\tau \circ \gamma_n - \tau \circ \gamma|^p d\mu$$

Given $x \in (\Omega \setminus E_\delta) \setminus g_\gamma^{-1}(A(\tau, 2\delta))$, we see that $|g_{\gamma_n}(x) - g_\gamma(x)| < \delta$ a.e, and $g_\gamma(x) \in \Omega \setminus A(\tau, 2\delta)$ so that $d(g_\gamma(x), \Omega \setminus S(\tau)) \geq 2\delta$ and thus $d(g_{\gamma_n}(x), \Omega \setminus S(\tau)) \geq \delta$.
This means that $g_{\gamma_n}(x)$ and $g_\gamma(x)$ are in the same connected component of $S(\tau)$ so that $|g_\tau \circ g_{\gamma_n}(x) - g_\tau \circ g_\gamma(x)| = |g_{\gamma_n} - g_\gamma(x)| < \delta$.

$$\implies I \leq \mu((\Omega \setminus E_\delta) \setminus g_\gamma^{-1}(A(\tau, 2\delta)))\delta^p + \mu((\Omega \setminus E_\delta) \cap g_\gamma^{-1}(A(\tau, 2\delta))) \operatorname{diam}(\Omega)^p$$
$$\leq \mu(\Omega)\delta^p + \mu(A(\tau, 2\delta)) \operatorname{diam}(\Omega)^p$$
$$\implies ||\tau \circ \gamma_n - \tau \circ \gamma||_{L^p(\Omega)} \leq \delta^{-p}||\gamma_n - \gamma||_{L^p(\Omega)} \operatorname{diam}(\Omega)^p + \mu(\Omega)\delta^p + \mu(A(\tau, 2\delta)) \operatorname{diam}(\Omega)^p$$

Now, choose $\delta = ||\gamma_n - \gamma||^{1/p+1}_{L^p(\Omega)}$.

$$\leq ||\gamma_n - \gamma||^{1/(p+1)}_{L^p(\Omega)} \operatorname{diam}(\Omega)^p + \mu(\Omega)||\gamma_n - \gamma||^{1/p+1}_{L^p(\Omega)} + \mu(A(\tau, 2||\gamma_n - \gamma||^{1/(p+1)})) \operatorname{diam}(\Omega)^p$$

This bound holds for all n. Now, we see that each term approaches 0 as $||\gamma_n - \gamma||$ does, so that $||\tau \circ \gamma_n - \tau \circ \gamma||_{L^p(\Omega)} \to 0$ as $||\gamma_n - \gamma||_{L^p(\Omega)} \to 0$ for $1 \leq p < \infty$.

Proposition 10 *If $X \subseteq \sigma(\Omega)$ is open under $||\cdot||_{L^p(\Omega)}$, then $\tau \circ X$ is as well for all $\tau \in \sigma(\Omega)$.*

Proof. Take any set X open in $||\cdot||_{L^p(\Omega)}$.
Now, take any sequence $\{\gamma_n\} \subseteq \sigma(\Omega) \setminus (\tau \circ X)$ convergent to some limit γ. We see that $\tau^{-1} \circ \gamma_n$ is then a sequence in $\sigma(\Omega) \setminus X$ convergent to $\tau^{-1} \circ \gamma$ by the previous proposition. Then, $\tau^{-1} \circ \gamma \in \Omega \setminus X$ as $\sigma(\Omega) \setminus X$ is closed. Finally, $\gamma \in \sigma(\Omega) \setminus (\tau \circ X)$, so that $\sigma(\Omega) \setminus (\tau \circ X)$ is closed and therefore $\tau \circ X$ is open.

Proposition 11 *$(\sigma(\Omega), ||\cdot||_{L^p(\Omega)})$ is a topological group.*

Proof. Assume that $||\tau_n - \tau||_{L^p} \to 0$. Now, we see that $||\tau^{-1} - \tau_n^{-1}||_{L^p(\Omega} = ||\tau^{-1} \circ \tau_n - \operatorname{Id}||_{L^p(\Omega)} = ||\tau^{-1} \circ \tau_n - \tau^{-1} \circ \tau||_{L^p(\Omega)} \to 0$ as well, so that $\tau \mapsto \tau^{-1}$ is continuous.
Now, assume that $||\tau_n - \tau||_{L^p} \to 0$ and $||\gamma_n - \gamma||_{L^p} \to 0$. We see that $||\tau_n \circ \gamma_n - \tau \circ \gamma||_{L^p(\Omega)} \leq ||\tau_n \circ \gamma_n - \tau \circ \gamma_n||_{L^p(\Omega)} + ||\tau \circ \gamma_n - \tau \circ \gamma||_{L^p(\Omega)} = ||\tau_n - \tau||_{L^p(\Omega)} + ||\tau \circ \gamma_n - \tau \circ \gamma||_{L^p(\Omega)}$. As both sequences approach 0, we see that $||\tau_n \circ \gamma_n - \tau \circ \gamma||_{L^p(\Omega)} \to 0$ as well, so that $(\tau, \gamma) \mapsto \tau \circ \gamma$ is continuous too.

Therefore, $(\sigma(\Omega), ||\cdot||_{L^p(\Omega)})$ is a topological group.

Now, the continuity of the group operations have been observed. As mentioned earlier, the compactness of subsets of $\sigma(\Omega)$ under the L^p norm is expected to be governed mostly by the sets of discontinuity of each of the elements. Accordingly, we prove the following theorem for relative compactness.

Proposition 12 *For $X \subseteq \sigma(\Omega)$, if $\mu(A(\tau, \varepsilon)) \to 0$ uniformly on X as $\varepsilon \to 0$, then X is relatively compact under L^p for $1 \leq p < \infty$.*

Proof. Pick any arbitrary p and choose $q < p^* = np/(n-p)$.
Let n^ε denote the usual mollifier taken by dilating n, a positive smooth function such that $\int_{\mathbb{R}^n} n d\mu = 1$ and 0 outside of $B(0,1)$. Take any sequence $\{\tau_n\} \subset X$ and let $\tau_n^\varepsilon = n^\varepsilon * \tau_n$.
First, we bound $||\tau_n^\varepsilon - \tau_n||_{L^q(\Omega)}$.

$$||\tau_n^\varepsilon - \tau_n||_{L^1(\Omega)} \leq \int_\Omega \int_{B(0,1)} n(y)|\tau_n(x-\varepsilon y) - \tau_n(x)|d\mu_y d\mu$$
$$\leq \int_{A(\tau,\varepsilon)} \int_{B(0,1)} n(y)|\tau_n(x-\varepsilon y) - \tau_n(x)|d\mu_y d\mu + \int_{B(0,1)} \int_{\Omega\setminus A(\tau,\varepsilon)} n(y)\varepsilon d\mu d\mu_y$$
$$\leq \int_{A(\tau,\varepsilon)} \int_{B(0,1)} n(y)|\tau_n(x-\varepsilon y) - \tau_n(x)|d\mu_y d\mu + \varepsilon\mu(\Omega)$$
$$\leq \mu(A(\tau,\varepsilon))\mu(B(0,1))\operatorname{diam}(\Omega) + \varepsilon\mu(\Omega)$$

As ε approaches 0^+, this upper bound approaches 0 by uniformly in n by hypothesis. Then, the rest of this proof follows exactly the same as Evan's proof of the Rellich-Kondrachov Compactness Theorem in 'Partial Differential Equations'.[**EV**]

Therefore, there must be some subsequence τ_{m_n} convergent in the $L^q(\Omega)$ norm so that X is relatively compact in $L^q(\Omega)$.

Now, for any arbitrary p, we can choose p_1 such that $p < p_1^*$, and thus X is relatively compact in $L^p(\Omega)$.

Next, we have seen that left composition by $\tau \in \sigma(\Omega)$ does not affect the L^p convergence of a sequence $\gamma_n \to \gamma$ in $\sigma(\Omega)$. We are able to prove an even stronger result as well.

Proposition 13 *Take any $1 \leq p, q < \infty$. For all $f \in L^q(\Omega)$, if $||\gamma_n - \gamma||_{L^p(\Omega)} \to 0$ then $||f\circ\gamma_n - f\circ\gamma||_{L^q(\Omega)} \to 0$.*

Proof. We first bound $||f\circ\gamma_n - f\circ\gamma||_{L^q(\Omega)}$. Choose any continuous and bounded function $r : \Omega \to \mathbb{R}$ such that $||f - r||_{L^q(\Omega)} < \delta_1$. Next, choose any closed set $A \subseteq \Omega$ with $\mu(\Omega \setminus A) < \delta_2$. In addition, since r is then uniformly continuous on A because A is compact, choose δ_3 so that $|x-y| < \delta_3 \implies |r(x) - r(y)| < \delta_4$ on E. In addition, let $E = \{x \in \Omega \mid |\gamma_n - \gamma| \geq \delta_3\}$ so that $\mu(E) \leq \delta_3^{-p}||\gamma_n - \gamma||^p_{L^p(\Omega)}$.
Then, we have the following.

$$||f\circ\gamma_n - f\circ\gamma||_{L^q(\Omega)} \leq ||f\circ\gamma_n - r\circ\gamma_n||_{L^q(\Omega)} + ||r\circ\gamma_n - r\circ\gamma||_{L^q(\Omega)} + ||r\circ\gamma - f\circ\gamma||_{L^q(\Omega)}$$
$$< ||r\circ\gamma_n - r\circ\gamma||_{L^q(\Omega)} + 2\delta_1$$
$$||r\circ\gamma_n - r\circ\gamma||^q_{L^q(\Omega)} = \int_{\Omega\setminus g_\gamma^{-1}(A)} |r\circ\gamma_n - r\circ\gamma|^q d\mu + \int_{g_\gamma^{-1}(A)} |r\circ\gamma_n - r\circ\gamma|^q d\mu$$

$$\leq (4\sup_{x\in\Omega}|r(x)|^q)\mu(g_\gamma^{-1}(\Omega\setminus A)) + \int_{g_\gamma^{-1}(A)\setminus E} |r\circ\gamma_n - r\circ\gamma|^q d\mu + \int_{g_\gamma^{-1}(A)\cap E} |r\circ\gamma_n - r\circ\gamma|^q d\mu$$
$$\leq (4\sup_{x\in\Omega}|r(x)|^q)(\mu(\Omega\setminus A) + \mu(E)) + \int_{g_\gamma^{-1}(A)\setminus E} |r\circ\gamma_n - r\circ\gamma|^q d\mu$$
$$\leq (4\sup_{x\in\Omega}|r(x)|^q)(\delta_2 + \delta_3^{-p}||\gamma_n - \gamma||^p_{L^p(\Omega)}) + \int_{(g_\gamma^{-1}(A)\setminus E)\cap g_{\gamma_n}^{-1}(A)} |r\circ\gamma_n - r\circ\gamma|^q d\mu + \int_{(g_\gamma^{-1}(A)\setminus E)\setminus g_{\gamma_n}^{-1}(A)} |r\circ\gamma_n - r\circ\gamma|^q d\mu$$

For $x \in (g_\gamma^{-1}(A) \setminus E) \cap g_{\gamma_n}^{-1}(A)$ we have that $g_\gamma(x), g_{\gamma_n}(x) \in A$ and $|g_\gamma(x) - g_{\gamma_n}(x)| < \delta_3$ a.e.so that $|r\circ\gamma_n - r\circ\gamma| < \delta_4$ a.e.

$$\leq (4\sup_{x\in\Omega}|r(x)|^q)(\delta_2 + \delta_3^{-p}||\gamma_n - \gamma||^p_{L^p(\Omega)}) + \int_{(g_\gamma^{-1}(A)\setminus E)\cap g_{\gamma_n}^{-1}(A)} \delta_4^q d\mu + \int_{\Omega\setminus g_{\gamma_n}^{-1}(A)} |r\circ\gamma_n - r\circ\gamma|^q d\mu$$
$$\leq (4\sup_{x\in\Omega}|r(x)|^q)(2\delta_2 + \delta_3^{-p}||\gamma_n - \gamma||^p_{L^p(\Omega)}) + \mu(\Omega)\delta_4^q$$

Note that these hold regardless of the $\delta_1, \delta_2, \delta_4 > 0$ chosen and so long as δ_3 is sufficiently small (with upper bound dependent upon δ_1 and δ_4).
Fix any $\varepsilon > 0$, and for any $\varepsilon > 0$ choose $\delta_1 > 0$ (and thus r) with $\delta_1 < \varepsilon/2$. Next, choose δ_4 (and thus δ_3) with $\delta_4 < \varepsilon^{1/q}\mu(\Omega)^{-1/q}$ and $\delta_2 < \varepsilon/(8\sup_{x\in\Omega}|r(x)|^q)$ so that we have the following.

$$||f\circ\gamma_n - f\circ\gamma||_{L^q(\Omega)} < \left(2\varepsilon + \delta_3^{-p}(4\sup_{x\in\Omega}|r(x)|^q)||\gamma_n - \gamma||_{L^p(\Omega)}\right)^{1/q} + \varepsilon$$

Finally, we may choose N such that $n > N \implies \delta_3^{-p}(4\sup_{x\in\Omega}|r(x)|^q)||\gamma_n - \gamma||_{L^p(\Omega)} < \varepsilon$, so that we have that $n > N$ implies the following.

$$||f\circ\gamma_n - f\circ\gamma||_{L^q(\Omega)} < (3\varepsilon)^{1/q} + \varepsilon$$

Then, as $\varepsilon > 0$ was arbitrary we have that $\lim_{n\to\infty}||f\circ\gamma_n - f\circ\gamma||_{L^q(\Omega)} \to 0$.

We have seen that L^p compactness is much simpler on $\sigma(\Omega)$, as it is governed mostly by the size of the sets of discontinuity. We see that in L^2 there is an even nicer characterization of compactness, due to the fact that L^2 is induced by an inner product.

Definition 6 *We define $\mathcal{T}_w$ to be the topology on $\sigma(\Omega)$ such that $\tau_n \to \tau$ if and only if $\langle\tau_n, f\rangle \to \langle\tau, f\rangle$ for all $f \in L^2(\Omega)$.*

Proposition 14 *$\mathcal{T}_w$ is equivalent to the L^2 topology on $\sigma(\Omega)$.*

Proof. It's clear that L^2 convergence implies convergence in $\mathcal{T}_w$. For the other direction, assume $\tau_n \to \tau$ in $\mathcal{T}_w$.

$$\begin{aligned}
\implies \langle\tau_n,\tau\rangle &= \frac{1}{2}(||\tau_n||^2_{L^2(\Omega)} + ||\tau||^2_{L^2(\Omega)} - ||\tau_n - \tau||^2_{L^2(\Omega)}) \\
&= ||\operatorname{Id}||^2_{L^2(\Omega)} - \frac{1}{2}||\tau_n - \tau||^2_{L^2(\Omega)} \\
&\to \langle\tau,\tau\rangle = ||\tau||^2_{L^2(\Omega)} \\
&= ||\operatorname{Id}||^2_{L^2(\Omega)} \\
\implies ||\tau_n - \tau||_{L^2(\Omega)} &\to 0
\end{aligned}$$

Therefore $\tau_n \to \tau$ in $\mathcal{T}_w$ if and only if $||\tau_n - \tau||_{L^2(\Omega)} \to 0$.

Proposition 15 *$(\sigma(\Omega), ||\cdot||_{L^2(\Omega)})$ is a compact topological group.*

Proof. It has already been proven that it is a topological group.
Because the weak and weak-* topologies coincide on L^2, we have by the Banach-Alaoglu Theorem [**LB**] that $B = \{f \in L^2 \mid \sup_{||g||_{L^2(\Omega)}<1/||\operatorname{Id}||_{L^2(\Omega)}}\langle f, g\rangle \le 1\})$ is compact in the weak topology.
We see that $\sigma(\Omega) \subseteq B$ so that $\sigma(\Omega)$ has compact closure in the weak-topology. Finally, since $\mathcal{T}_w$ is the subspace topology inherited from the weak topology on L^2, $(\sigma(\Omega), \mathcal{T}_w) = (\sigma(\Omega), ||\cdot||_{L^2(\Omega)}\rangle$ is compact.

Definition 7 *As $(\sigma(\Omega), ||\cdot||_{L^2(\Omega)})$ is a compact topological group, there must exist some right-invariant measure on subsets of $\sigma(\Omega)$, denoted $\mu_{\mathbb{D}}$, and normalized so that $\mu_{\mathbb{D}}(\sigma(\Omega)) = 1$.*

5 Algebraic Properties

Next, each τ induces an isometry in L^p by the mapping $f \mapsto f\circ\tau$. This begs the question of which isometries are induced by $\tau \in \sigma(\Omega)$ in some form.

Proposition 16 *For any linear map $M : L^2(\Omega,\mathbb{R}) \to L^2(\Omega,\mathbb{R})$ which preserve $||\cdot||_{L^1(\Omega,\mathbb{R})}$, $||\cdot||_{L^\infty(\Omega,\mathbb{R})}$, $\langle\cdot,\cdot\rangle$, and such that $f \ge 0 \implies Mf \ge 0$, we have that $Mf = f\circ\gamma^{-1}$ for $\gamma \in \sigma(\Omega)$.*

Proof. Take such an M. It preserves $||\cdot||_{L^2(\Omega,\mathbb{R})}$ as well since it preserves inner products.
Now, take any measurable set $A \in \Sigma_\Omega$ with nonzero measure and let χ_A be its indicator function. We see that $0 \leq M\chi_A \leq 1$ almost everywhere since $||M\chi_A||_{L^{\infty,\mathbb{R}}} = 1$.
Assume for a contradiction that $K = \{x \in \Omega \mid 0 < M\chi_A < 1\}$ has nonzero measure. Then, we have the following.

$$\begin{aligned}
&\mu(A) = \langle M\chi_A, M\chi_A\rangle \\
&= \int_{\Omega\setminus K} (M\chi_A)^2 d\mu + \int_K (M\chi_A)^2 d\mu \\
&= \int_{\Omega\setminus K} M\chi_A d\mu + \int_K (M\chi_A)^2 d\mu \\
&< \int_{\Omega\setminus K} M\chi_A d\mu + \int_K M\chi_A d\mu \\
&= \int_\Omega |M\chi_A| d\mu \\
&= ||M\chi_A||_{L^1(\Omega,\mathbb{R})} = \mu(A)
\end{aligned}$$

Then, we have a contradiction, so that $\mu(K) = 0$ and thus $M\chi_A \in \{0, 1\}$. Now, we may define $\tau : \Sigma_\Omega \to \Sigma_\Omega$ by $\tau(A) = \{x \in \Omega \mid M\chi_A = 1\}$. Note that we now have that $\mu(A) = ||M\chi_A||_{L^1(\Omega,\mathbb{R})} = ||\chi_{\tau(A)}||_{L^1(\Omega,\mathbb{R})} = \mu(\tau(A))$.
For disjoint A, B we have by the linearity of M that $M\chi_{A\cup B} \equiv M\chi_A + M\chi_B \equiv \chi_{\tau(A)} + \chi_{\tau(B)}$ so that $\tau(A), \tau(B)$ are disjoint as well (modulo a null set) and thus $\tau(A \cup B) = \tau(A) \cup \tau(B)$ (again modulo a null set).
The same logic shows that $\tau(A \setminus B) = \tau(A) \setminus \tau(B)$.
Thus, by Theorem 9.5.1 in [**BG**] we have that there exists some $\gamma : \Omega \to \Omega$ such that $\mu(\tau(A)\Delta\gamma(A)) = 0$ for all $A \in \Sigma_\Omega$.
Then, $|M\chi_A| \equiv \chi_{\gamma(A)} \equiv \chi_A \circ \gamma^{-1}$, so that $|Mf| \equiv f \circ \gamma^{-1}$ for simple functions $f \in L^\infty(\Omega, \mathbb{R})$ as well, and therefore $|Mf| \equiv f \circ \gamma^{-1}$ for all $f \in L^\infty(\Omega, \mathbb{R})$ by taking limits of simple functions.

Definition 8 *We denote the set of linear maps M as in the hypotheses of the previous proposition to be $L(\Omega)$, the linear representation of Ω. We have that $L(\Omega)$ is a subgroup of the group $G(\Omega) = U(L^2(\Omega, \mathbb{R})) \cap U(L^1(\Omega, \mathbb{R})) \cap U(L^\infty(\Omega, \mathbb{R}))$ (with the group action being multiplication/composition).*

Even if $M \in G(\Omega) \setminus L(\Omega)$, we still see that M must be induced (in a way) by some $\tau \in \sigma(\Omega)$.

Proposition 17 *For all $M \in G(\Omega)$, we have that $M = \lambda N$ for $\lambda \in L^\infty(\Omega, \mathbb{R})$ with $\lambda = \pm 1$ a.e., and $N \in L(\Omega)$.*

Proof. Take any $M \in G(\Omega)$. Take any measurable $A \in \Sigma_\Omega$, so we see that $\mu(A) = ||M\chi_A||_{L^1(\Omega,\mathbb{R})}$.
Then, consider $M\chi_\Omega$. If $|M\chi_\Omega| < 1$ on a set T of nonzero measure then since $||M\chi_\Omega||_{L^\infty(\Omega,\mathbb{R})} = 1$ we have that $\mu(\Omega) = \int_\Omega |M\chi_\Omega| d\mu = \int_T |M\chi_\Omega| d\mu + \int_{\Omega\setminus T} |M\chi_\Omega| d\mu < \mu(T) + \mu(\Omega \setminus T) = \mu(\Omega)$, a contradiction. Then, $|M\chi_\Omega| = 1$ almost everywhere so that $(M\chi_\Omega)M \in G(\Omega)$ as well.

Define $K = \{x \in \Omega \mid (M\chi_\Omega)(M\chi_A) < 0\}$. If $x \in K$ then since $|(M\chi_\Omega)(M\chi_A)| = 0$ or 1 we have that $(M\chi_\Omega)(M\chi_A) = -1$. Then, $\langle (M\chi_\Omega)(M\chi_A), \chi_K\rangle = -\mu(K)$. Next, we have the following.

$$\begin{aligned}
\langle M\chi_\Omega M\chi_A, 1\rangle &= \langle M\chi_\Omega M\chi_A, \chi_K\rangle + \langle M\chi_\Omega M\chi_A, \chi_{\Omega\setminus K}\rangle \\
&\leq -\mu(K) + ||M\chi_\Omega M\chi_A||_{L^1(\Omega,\mathbb{R})} ||\chi_{\Omega\setminus K}||_{L^\infty(\Omega,\mathbb{R})} \\
&= \mu(A) - \mu(K)
\end{aligned}$$

However, we also have that $\langle M\chi_\Omega M\chi_A, 1\rangle = \langle M\chi_\Omega, M\chi_A\rangle = \langle \chi_\Omega, \chi_A\rangle = \mu(A)$. Therefore, since $\mu(A) \leq \mu(A) - \mu(K)$, we have that $\mu(K) = 0$, and that $M\chi_\Omega M\chi_A \geq 0$ a.e.
Then, $M\chi_\Omega Mf \geq 0$ for all simple functions $f \geq 0$, and thus by taking limits we see that $M\chi_\Omega Mf \geq 0$ for all $f \in L^1(\Omega, \mathbb{R})$ which are ≥ 0.

Finally, we have shown that $(M\chi_\Omega)M \in L(\Omega)$ and $M\chi_\Omega = \pm 1$ almost everywhere.

Corollary. Any $N \in G(\Omega)$ is thus also in $U(L^p(\Omega,\mathbb{R}))$ for all other $1 \leq p < \infty$, since $Nf = (N\chi_\Omega)f \circ \tau^{-1}$ for some $\tau \in \sigma(\Omega)$ so that $||Nf||^p_{L^p(\Omega,\mathbb{R})} = \int_\Omega |(N\chi_\Omega)f \circ \tau^{-1}|^p d\mu = \int_\Omega |f \circ \tau^{-1}|^p d\mu = ||f||^p_{L^p(\Omega)}$.
Therefore, with $U(L^p(\Omega,\mathbb{R}))$ the group of $||\cdot||_{L^p(\Omega)}$ preserving linear transformations (with the operation being composition), we have that $\cap_{p\in[1,\infty]} U(L^p(\Omega,\mathbb{R})) = U(L^1(\Omega,\mathbb{R})) \cap U(L^2(\Omega,\mathbb{R})) \cap U(L^\infty(\Omega,\mathbb{R}))$.

Proposition 18 *The homomorphism $\zeta : \sigma(\Omega) \to L(\Omega)$ given by $\zeta(\tau)f = f \circ \tau^{-1}$ is a homeomorphism when $L(\Omega)$ is given the strong/pointwise topology and $\sigma(\Omega)$ is given the $||\cdot||_{L^p(\Omega)}$ topology.*

Proof. Take any sequence $\{\gamma_n\} \subset \sigma(\Omega)$ convergent in $||\cdot||_{L^p(\Omega)}$ to some $\gamma \in \sigma(\Omega)$. Now, take any $f \in L^2(\Omega)$. Because $||\gamma_n - \gamma||_{L^p(\Omega)} \to 0$ and $(\sigma(\Omega), ||\cdot||_{L^p(\Omega)})$ is a topological group we see that $||\gamma_n^{-1} - \gamma^{-1}||_{L^p(\Omega)} \to 0$ as well so that by the previous proposition we have that $||f \circ \gamma_n^{-1} - f \circ \gamma||_{L^2(\Omega)} \to 0$ so that $\zeta(\gamma_n)f \to \zeta(\gamma)f$ and thus $\zeta(\gamma_n) \to \zeta(\gamma)$ in the strong/pointwise topology on $U(L^2(\Omega))$.

Conversely, assume that $\zeta(\gamma_n) \to \zeta(\gamma)$ in the strong/pointwise topology on $L(\Omega)$. We see that $||\operatorname{Id} \circ \gamma_n^{-1} - \operatorname{Id} \circ \gamma^{-1}||_{L^2(\Omega)} \to 0$ so that $||\gamma_n - \gamma||_{L^2(\Omega)} \to 0$ as well since $\sigma(\Omega)$ is a topological group. Then, $||\gamma_n - \gamma||_{L^1(\Omega)} = \int_\Omega |\gamma_n - \gamma| d\mu \leq ||1||_{L^2(\Omega)} ||\gamma_n - \gamma||_{L^2(\Omega)} \to 0$, so that $||\gamma_n - \gamma||_{L^p(\Omega)} \to 0$ for all $p \geq 1$. Finally, this implies that $||\gamma_n - \gamma||_{L^p(\Omega)} \to 0$ as well for all $p \geq 1$, so that $\zeta(\gamma_n)$ converges if and only if γ_n converges in $||\cdot||_{L^p(\Omega)}$ so that $\sigma(\Omega)$ is homeomorphic to $\zeta(\sigma(\Omega)) = L(\Omega)$.

Proposition 19 *The center Z of $\sigma(\Omega)$ is precisely* Id.

Proof. Assume $\tau \in Z$, and assume for a contradiction that for some $A \subseteq S(\tau)$ we have that $g_\tau(A) \neq A$ and $\mu(A) > 0$. Then, there must exist some open $C \subseteq A$ such that $g_\tau(C) \cap C = \emptyset$ and $\mu(C) > 0$.
Now, there must be some $B(y,r) \subseteq C$ so that we may take $C = B(y,r)$. Because g_τ is continuous on C we have that $g_\tau(B(y,r)) = B(g_\tau(y),r)$. We can take $r > 0$ small enough so that $B(g_\tau(y),r) \cap B(y,r) = \emptyset$.
Let $B(z,r) \subseteq S(\tau)$ be another set disjoint from both $B(y,r)$ and $B(g_\tau(y),r)$, and let γ be the permutation with $\gamma(x) = x + z - g_\tau(y)$ for $x \in B(g_\tau(y),r)$, $\gamma(x) = x + g_\tau(y) - z$ for $x \in B(z,r)$, and $\gamma(x) = x$ otherwise.
We see that $g_\tau(\gamma(B(y,r))) = g_\tau(B(y,r)) = B(g_\tau(y),r)$ and $\gamma(g_\tau(B(y,r))) = \gamma(B(g_\tau(y),r)) = B(z,r)$.
Therefore, we see that $g_\tau \circ \gamma \neq \gamma \circ g_\tau$, a contradiction to the fact that $g_\tau \in Z$ (which must be true because $\tau \in Z$).
Therefore, $g_\tau(A) = A$ for all $A \subseteq \Omega$ with $\mu(A) \geq 0$ so that τ is equivalent a.e. to Id.

6 Future Directions

In some sense, functions in $\sigma(\Omega)$ are closely analogous to permutations $p \in S_n$. For example, $\sum_{m=1}^n f(m) = \sum_{m=1}^n f(p(m))$ is analogous to $\int_\Omega f d\mu = \int_\Omega f \circ \tau d\mu$. In addition, $\{1,\ldots,n\}$ can be partitioned into intervals $[a,b]$ such that p has slope ± 1, similar to how Ω can be partitioned into sets so that τ has a unitary slope.
Moreover, the problem originally arose when trying to generalize the finite-dimensional determinant to an infinite-dimensional one, with matrices being functions $A : \Omega^2 \to \mathbb{C}$ instead of $A : \{1,\ldots,n\}^2 \to \mathbb{C}$. Symbolically, the generalization using the Leibniz formula for the determinant is straightforward.

$$\sum_{p \in S_n} \operatorname{sgn}(p) \prod_i A_{ip(i)}$$

$$\int_{\sigma(\Omega)} \operatorname{sgn}(\tau) \exp\left(\int_\Omega \ln(A(x,\tau(x)))d\mu\right) d\mu_{\mathbb{D}}$$

However, proving properties about the determinant (even $\det(AB) = \det(A)\det(B)$ or some similar version) has proven difficult. Heuristically, the proof that $\det(AB) = \det(A)\det(B)$ using the Leibniz determinant involves a sum over $x \in \{1,\ldots,n\}^n$. The counting measure on $\{1,\ldots,n\}^n$ reduces to the counting measure on S_n when each element of S_n is identified with the corresponding element of $\{1,\ldots,n\}^n$, and so for the infinite case we would want a measure on Ω^Ω (or some similarly large set) which reduces to $\mu_{\mathbb{D}}$ on $\sigma(\Omega)$.

The natural log in the product integral also introduces issues stemming from the complex logarithm being multivalued. Using alternate product integrals, like the Volterra integral, has not yielded much more.

Care must be taken to define sgn appropriately as well. Attempts to define sgn have involved defining sgn as a homomorphism $\sigma(\Omega) \to GL(n, \mathbb{R})$. Fathi (1978) showed that $\sigma(\Omega)$ is a simple (and thus perfect) group, so that Im(sgn) cannot be abelian.

Further developments may prove the same results for a larger class of Ω including the finite sets $\{1, \ldots, n\}$ so that the analogy between $\sigma(\Omega)$ and the S_n can be made more rigorous. These may also allow a generalization of Leibniz' determinant formula to infinite-dimensional 'matrices' (functions $\Omega^2 \to \mathbb{C}$).